

FreeCAD for 3D Printing

PROJECT 05 OF 40

Custom Name Keychain Tag

Step-by-Step PDF Guide for Multi-Color 3D Printing



What's Included in This Package

 This Guide	Detailed step-by-step PDF tutorial with screenshots
 .FCStd File	FreeCAD project file — ready to open and modify
 .3MF File	Pre-sliced multi-color export, ready for Bambu Studio
 Font Bundle	Curated font collection — free for personal & commercial use

Get support & more resources at Ko-fi:

<https://ko-fi.com/cadcamcourse>

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Introduction

Welcome to the FreeCAD for 3D Printing series — a complete learning journey designed to take you from basic modelling to designing functional, ready-to-print parts.

In this project, we will model a custom name keychain tag that is fully ready for multi-color 3D printing. By the end of this tutorial, you will be able to create your own personalized keychains using any name or text you like.

About This Series

The FreeCAD for 3D Printing series covers 40 projects in total. Each project introduces new tools and techniques in a structured, beginner-friendly way.



Project Overview

Series	FreeCAD for 3D Printing
Project No.	05 of 40
Difficulty	Beginner — Intermediate
Workbenches Used	Draft → Part → Sketcher → Part
Output Format	.3MF (multi-color, multi-part compound)
Slicer Shown	Bambu Studio (similar steps apply to other slicers)

What You Will Learn

- How to create a 3D text shape using the Draft Workbench Shape String tool
- How to extrude a 2D shape into a 3D solid in the Part Workbench
- How to apply a 2D Offset to create an expanded surrounding layer
- How to use FaceBinder to extract reusable 2D faces from a 3D object
- How to design and position a keychain ring/loop attachment
- How to use the Compound operation to prepare a model for multi-color printing
- How to assign colors to individual parts in Bambu Studio

Software & Files Required

FreeCAD	Version 1.0.0 or newer — freecad.org (free & open source)
Slicer	Bambu Studio / OrcaSlicer / PrusaSlicer (any .3MF-capable slicer)
Font File	Included in the font bundle (see package contents on cover)
Project File	Michael_Keychain.FCStd — included in the download package
3MF Export	Michael_Keychain.3mf — ready to open in your slicer

Version Note

This tutorial was created using FreeCAD 1.1.0. If you are on an older version, the steps will be very similar, but menu locations or icons may differ slightly.

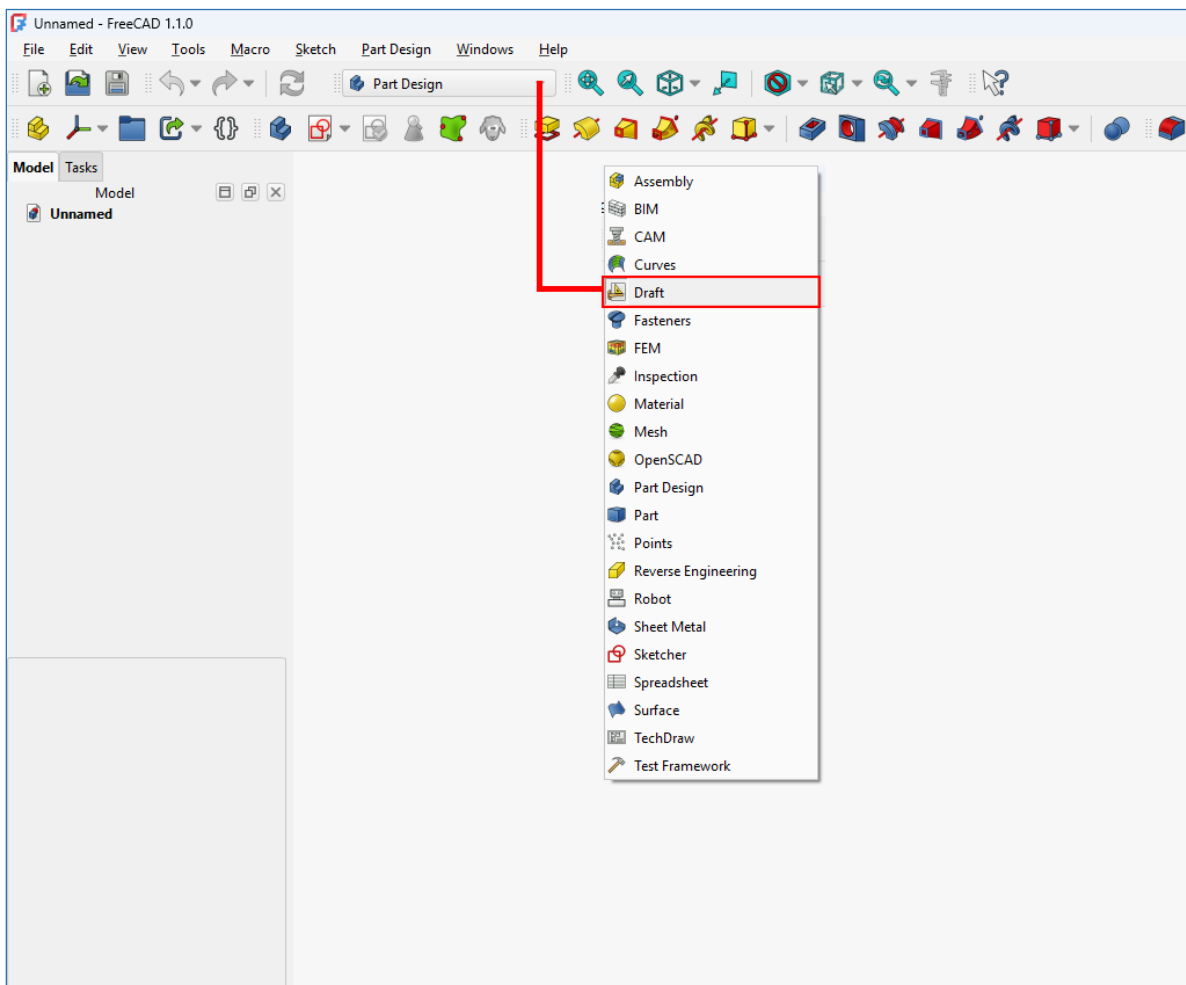
STEP 1 CREATE THE TEXT SHAPE (DRAFT WORKBENCH)

In this step, we use the Draft Workbench's Shape String tool to generate a 2D text outline from any font of your choice. This text shape will serve as the foundation for the entire model.

1.1 Switch to the Draft Workbench

FreeCAD starts in a default workbench. We need to switch to the Draft Workbench to access the Shape String tool.

1. Click the Workbench dropdown menu (top-left area of the screen).
2. Scroll through the list and select Draft .
3. The toolbar and menu items will update to reflect the Draft Workbench tools.



Tip

You can also switch workbenches by going to View → Workbench in the main menu.

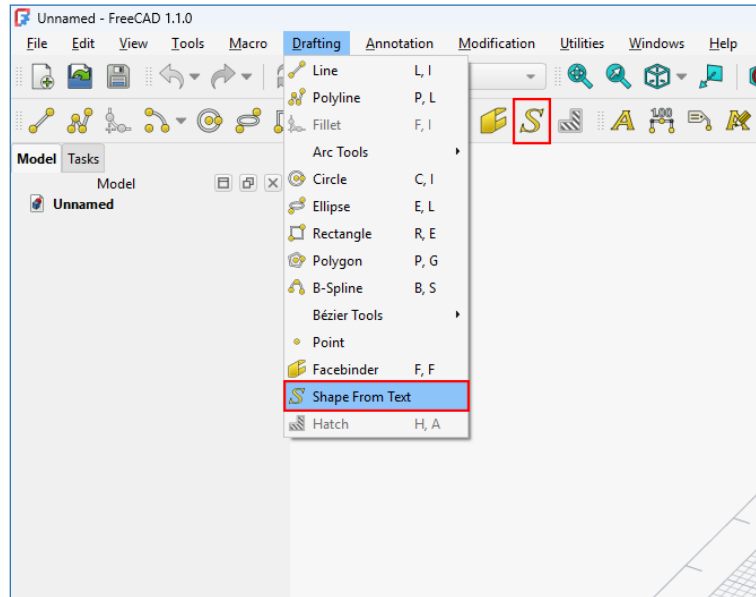
1.2 Open the Shape from Text Tool

The Shape String tool creates a 2D vector outline of any text using a specified font file.

Navigate to the tool using the menu path below:

► **Drafting** → **Shape from Text** *S*

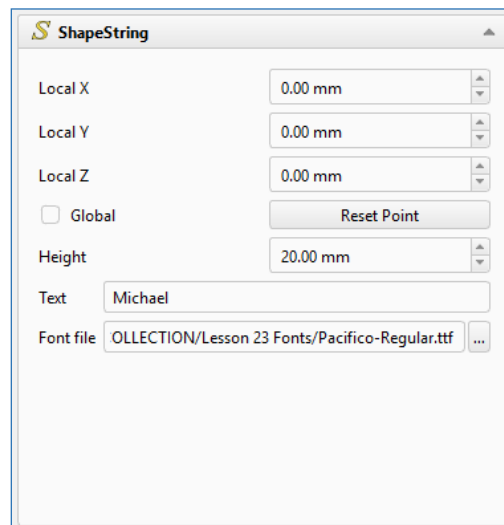
OR click the Shape from Text icon *S* directly in the Draft toolbar.



1.3 Configure the Shape String Settings

The Shape String dialog has several fields to configure. Use the table below as a reference:

Text	Enter your desired name or custom text (e.g., Michael)
Font File	Click the '...' button → browse to your font file (.ttf or .otf)
Height	Set to 20 mm (you can adjust this to suit your design)
X / Y / Z	Set all to 0, or leave as default — this does not affect the outcome

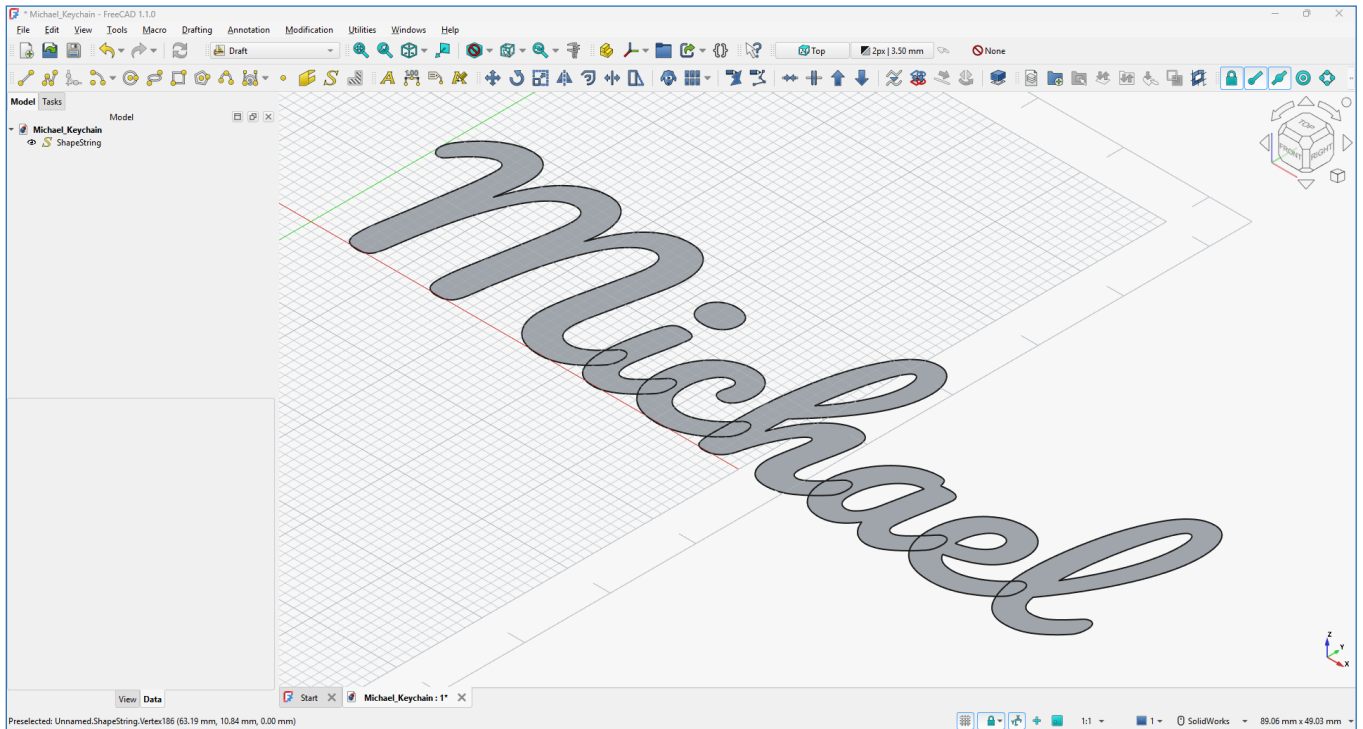


💡 Font Tips

This tutorial uses a cursive-style font. The included font bundle contains a curated selection of fonts free for personal and commercial use.

1.4 Accept

1. After entering your settings, Click OK to confirm and generate the text shape.



☑ Step 1 Complete

Your text shape is now created as a flat 2D object in the model tree (listed as ShapeString under the Draft objects). We will extrude this into a 3D solid in the next step.

STEP 2 EXTRUDE TEXT INTO A 3D SOLID (PART WORKBENCH)

Now we convert our flat 2D text shape into a proper 3D solid by extruding it upward. We also merge the individual letter shapes into one unified object.

2.1 Switch to the Part Workbench

The extrusion and Boolean operations we need are in the Part Workbench — not Part Design. Make sure you select the correct one.

1. Click the Workbench dropdown.
2. Select Part  (not Part Design .

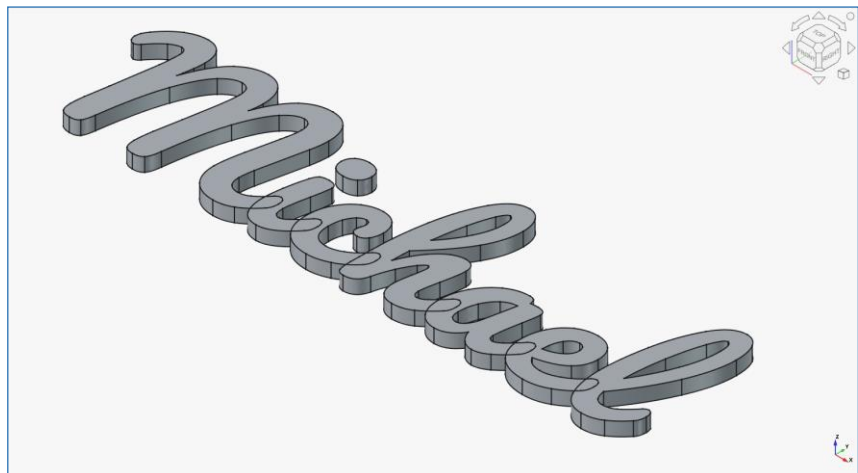
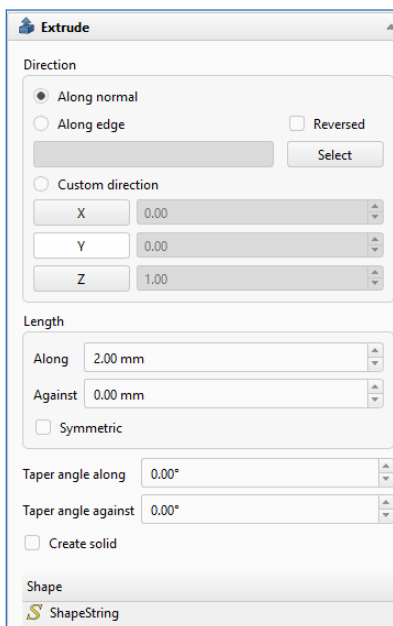
Important

Part Workbench and Part Design Workbench are different. For this workflow, always use the Part Workbench when working with Shape Strings and offsets.

2.2 Select and Extrude the Text Shape

1. In the Model Tree (left panel), click to select the ShapeString object.
2. Go to Part → Extrude , or click the Extrude icon  in the toolbar.
3. In the Extrude dialog, find the Length Along parameter.
4. Set the extrusion length to 2 mm.
5. Leave all other parameters at their default values.
6. Click OK.

► Part → Extrude 

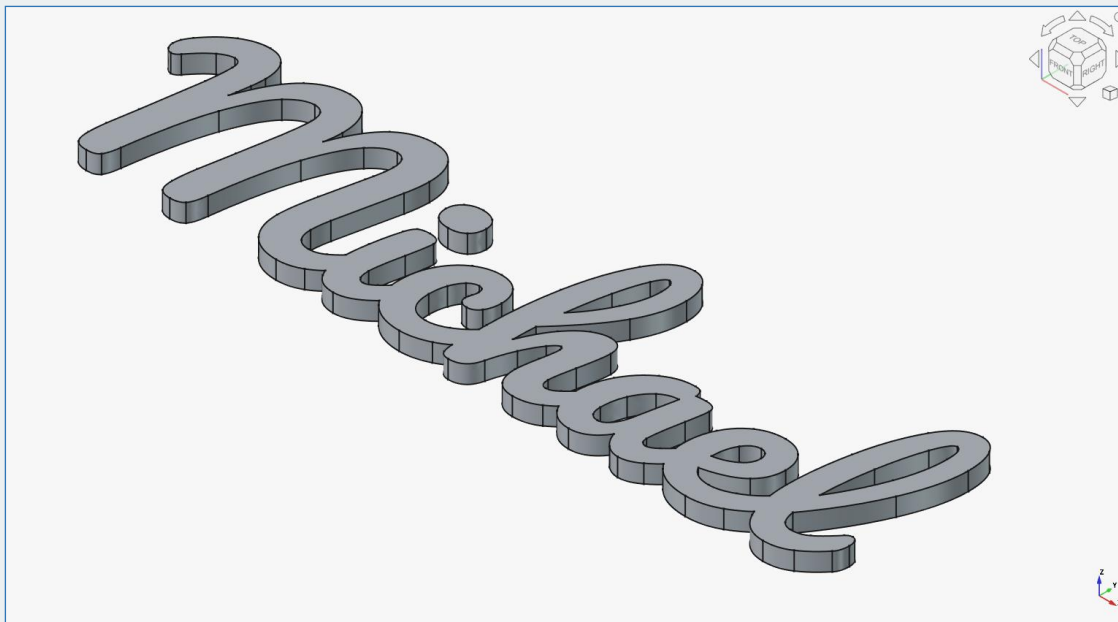


2.3 Merge Letters with Boolean Union

Even with a cursive font where letters appear connected, FreeCAD still treats each letter (and sub-element) as a separate solid. We need to merge them into one clean object.

1. Select the Extrusion object from the Model Tree.
2. Go to Part → Boolean Operations → Union , or click the Union icon  in the toolbar.
3. All connected letter parts will be fused into a single solid.

► Part → Boolean Operations → Union 



Union result — all connected letters merged into one solid.
Note: small separate elements (like the dot on 'i') may remain separate — this is normal.

Non-Cursive Fonts

If you chose a non-cursive (block) font where letters are not connected, each letter will remain a separate solid after Union. This is fine — the workflow is the same, and your slicer will handle them as individual parts in the multi-color step.

Step 2 Complete

The text is now a 3D solid. You can see it as a Fusion object in the Model Tree. This is the first (bottom) layer of the keychain.

STEP 3 CREATE THE SECOND LAYER WITH 2D OFFSET

The keychain has two layers — the text layer (Layer 1) and a slightly larger background plate (Layer 2). In this step, we derive Layer 2 by applying an outward offset to the text shape.

3.1 Extract a 2D Face from the 3D Solid

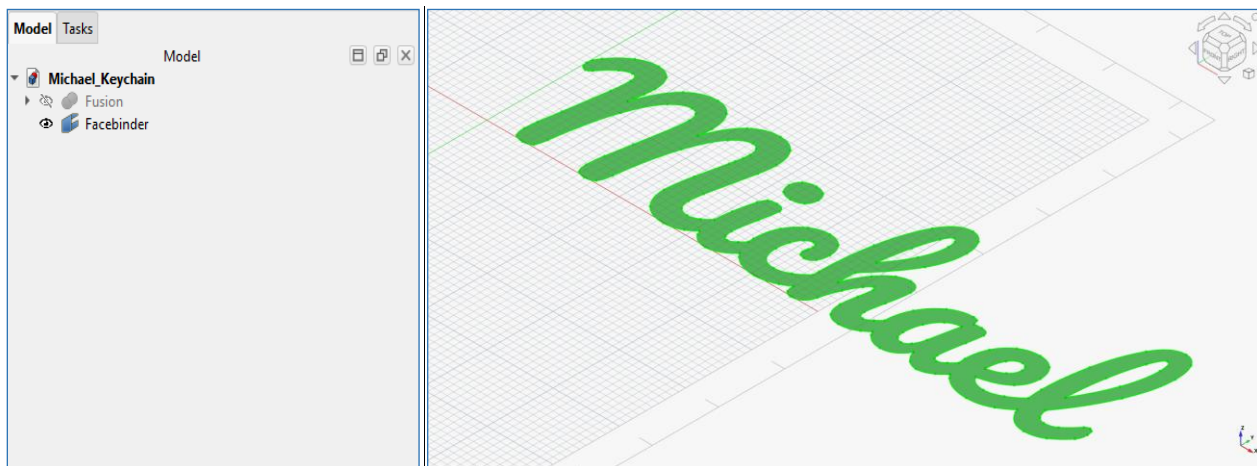
We cannot apply a 2D offset to a 3D object directly. We must first extract a flat 2D version of the text profile using the FaceBinder tool.

1. Rotate the viewport to clearly see the back face of the extruded text.
2. Hold the middle mouse button and drag, or use the navigation cube.
3. Hold Ctrl and click to select all back faces of the text solid.
4. Switch back to the Draft Workbench (Workbench dropdown → Draft).
5. Go to Drafting → Face Binder, or click the FaceBinder icon.

► Drafting → Face Binder



Back faces of the extruded text selected — viewport rotated



FaceBinder object created — Model Tree showing 'FaceBinder' entry

💡 Visibility Tip

After creating the FaceBinder, click the eye icon  next to the Extrude object in the Model Tree to hide it. This lets you clearly see the clean 2D FaceBinder shape.

3.2 Apply the 2D Offset

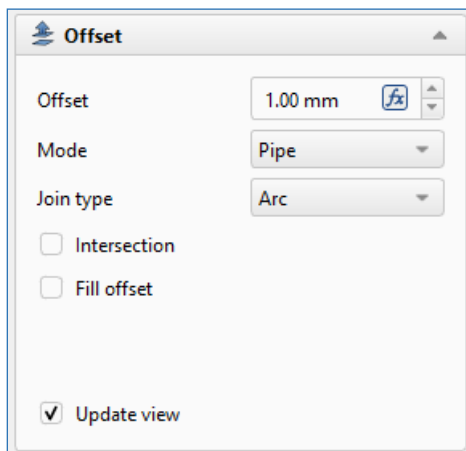
We now expand the FaceBinder shape outward by a small distance to create a slightly larger background layer.

1. Switch to the Part Workbench .
2. Select the FaceBinder object in the Model Tree.
3. Go to Part → 2D Offset , or click the 2D Offset icon .

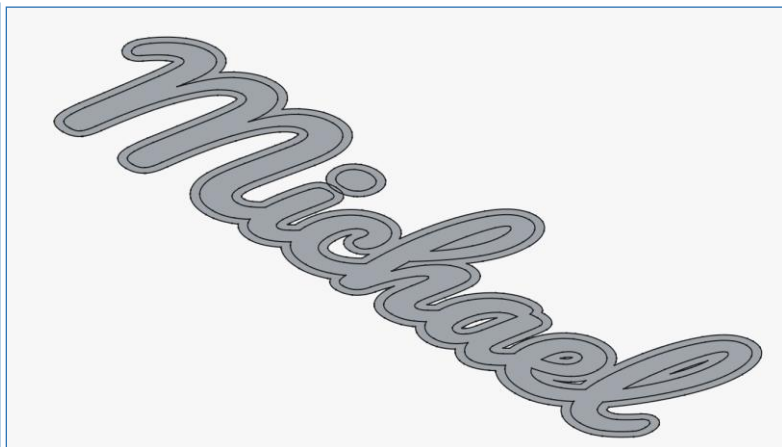
► Part → 2D Offset 

Offset Distance	1 mm (adjustable — larger value = thicker border)
Mode	Pipe
Join Type	Arc (gives smooth rounded corners)

4. A preview of the offset shape will appear in the viewport.
5. Verify the outline correctly wraps around all letters.
6. Click **OK** to confirm.



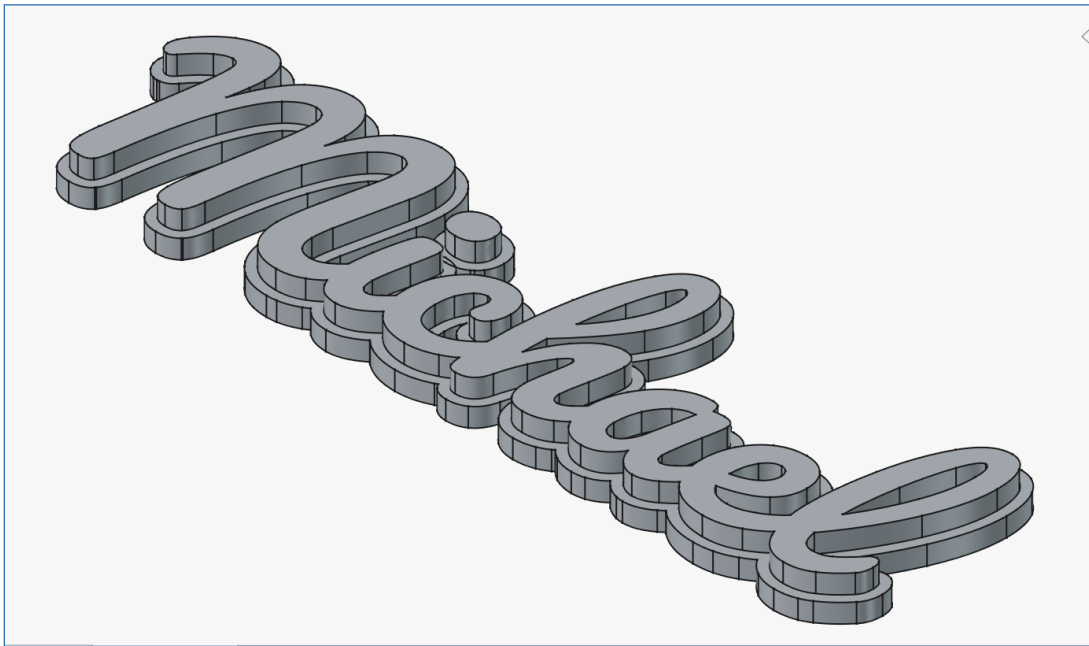
2D Offset dialog



Resulting offset shape in the viewport — expanded outline around the text

3.3 Extrude and Union the Second Layer

1. Hide the FaceBinder object (click the eye icon ) to see the offset shape clearly.
2. Select the Offset object from the Model Tree.
3. Apply the Extrude operation  (Part → Extrude ) .
4. Set the extrusion length to 2 mm (matching Layer 1).
5. Click OK.
6. Select the new Extrude from the Model Tree.
7. Apply the Union operation  (Part → Boolean Operations → Union ) to merge all segments into one solid.



☒ Step 3 Complete

Layer 2 is now a clean single solid that perfectly borders the text layer. Turn Layer 1 visibility back on — you should see both layers perfectly aligned on top of each other.

STEP 4 ADD THE KEYCHAIN RING LOOP

A keychain needs a ring to attach to keys. In this step, we sketch an annular (donut-shaped) loop directly on the surface of Layer 2 and fuse it into the model.

4.1 Prepare the Viewport

To work more clearly, hide Layer 1 temporarily so only the background plate (Layer 2) is visible.

1. Click the eye icon  next to the Layer 1 fusion object to hide it.

4.2 Create a Sketch on the Top Surface

1. Click the top face of the Layer 2 solid to select it.
2. Click the **Create New Sketch icon**  in the toolbar.
3. The Sketch Attachment dialog will appear.
4. In the dropdown, select **Plane Face**.
5. Click **OK**.

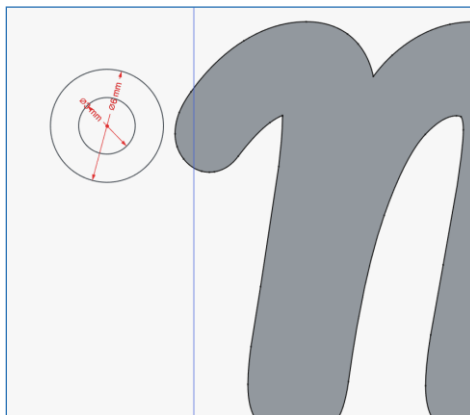
You are now inside the *Sketcher Workbench*  with the sketch attached to the top surface.

4.3 Draw the Concentric Ring

We will draw two concentric circles — the outer circle defines the ring's outer diameter, and the inner circle creates the hole for the keyring.

1. Activate the **Circle tool**  from the Sketcher toolbar.
2. Click near the left edge of the model to place the center point.
3. Drag outward and click to set the outer circle radius.
4. Apply a radius constraint: set the outer circle diameter to 6 mm.
5. Using the same center point, draw a second, smaller circle.
6. Apply a radius constraint: set the inner circle diameter to 3 mm.

Outer Circle Diameter	6 mm (ring outer edge)
Inner Circle Diameter	3 mm (hole for the keyring)
Wall Thickness	1.5 mm (calculated: $(6-3) / 2$)



Two concentric circles drawn in the Sketcher — both constrained, diameters labeled

4.4 Position the Loop

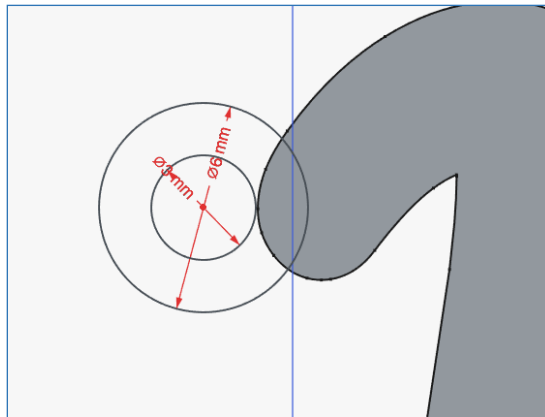
At this stage, the position of the ring is unconstrained (only the sizes are fixed). We position it by dragging.

1. Click and drag the center point of the circles.
2. Position the loop so it slightly overlaps the body of the keychain.

This overlap is critical — it creates the connection between the loop and the body when printed. The greater the overlap, the stronger the joint.

Placement Tip

Depending on the first letter of your text, the ideal overlap position may vary. For the name 'Michael', the left edge of the letter 'M' provides a natural attachment point. Try to overlap by at least 1–1.5 mm for strength.



Ring loop positioned overlapping the left edge of the keychain body

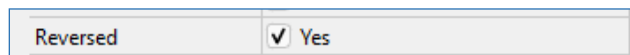
3. Once satisfied with the position, click Close Sketch (or press Esc and confirm).

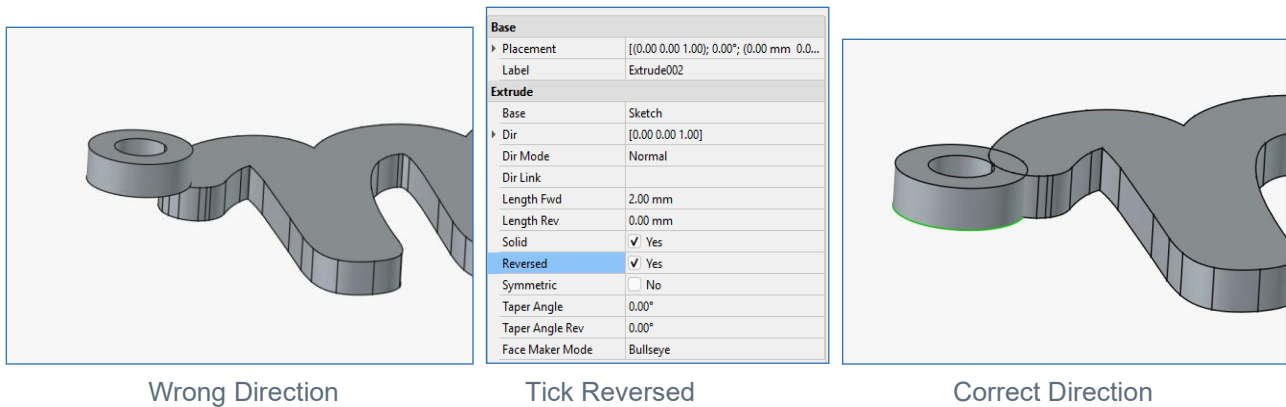
4.5 Extrude the Ring Loop

1. Switch to the Part Workbench  (Workbench dropdown → Part .
2. Select the Ring loop sketch from the Model Tree.
3. Apply the Extrude operation  (Part → Extrude .
4. Set the extrusion length to 2 mm (same as both layers).
5. Click OK.

If the extrusion goes in the wrong direction (below the plate instead of above):

6. Select the Extrude object in the Model Tree.
7. In the Data tab (Properties panel), find the Reversed field.
8. Check the checkbox to reverse the direction.

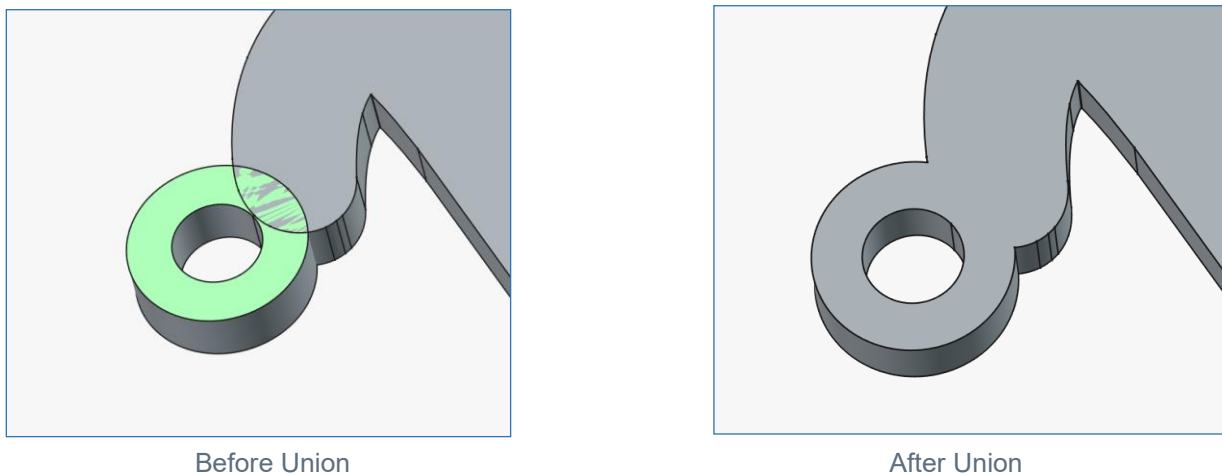




4.6 Fuse the Loop into Layer 2

1. Hold Ctrl and select both the ring loop extrusion and the Layer 2 fusion object.
2. Apply the Union operation (Part → Boolean Operations → Union).

Both parts merge into a single solid. The loop is now a permanent part of Layer 2.

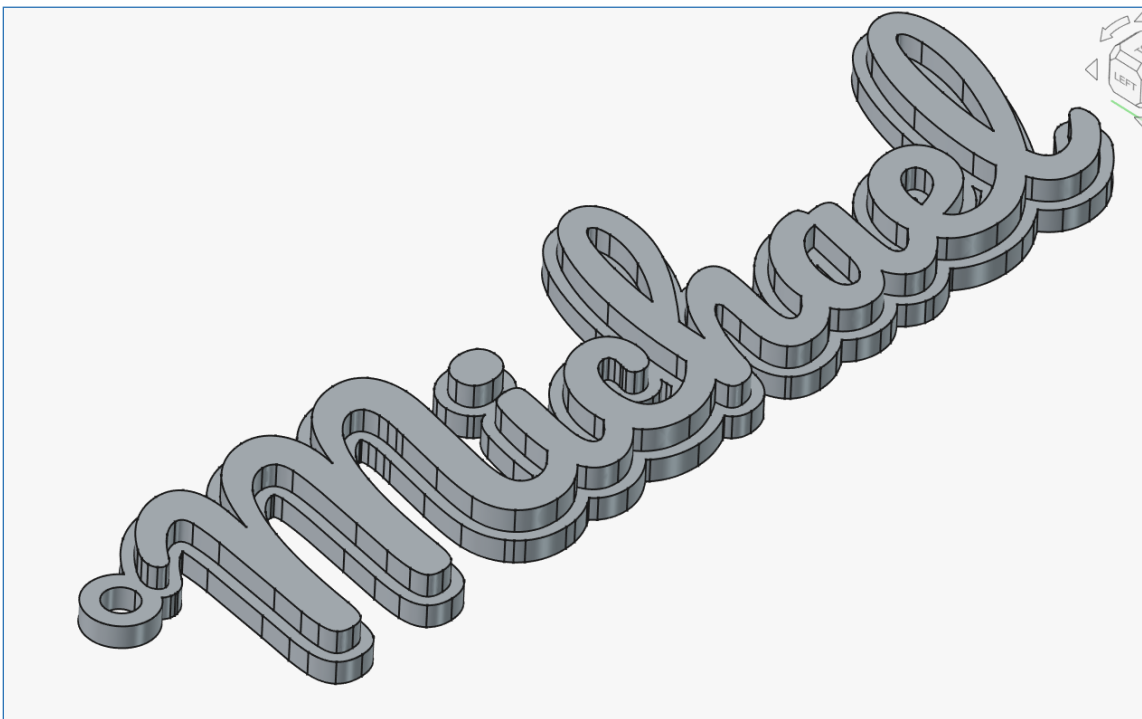


Editing the Loop Later

Because these operations are non-destructive, you can reposition the loop at any time. In the Model Tree, expand the Fusion → Extrude → Sketch, right-click the sketch, and choose Edit Sketch. Drag the center point to your new position and close the sketch. FreeCAD will automatically rebuild the model.

3. Turn Layer 1 visibility back on .

You should now see both layers — with the ring loop extending from the left edge.



Final model with both layers visible — ring loop on the left, letters on top

☒ Step 4 Complete

The keychain is fully modeled! You now have Layer 1 (text) and Layer 2 (background + loop) as two separate solid objects. In the final step, we prepare this for multi-color printing.

STEP 5 EXPORT FOR MULTI-COLOR 3D PRINTING

In this final step, we group both layers into a Compound object, export it as a .3MF file, and assign different filament colors in the slicer. The result is a print-ready two-color keychain.

5.1 Create a Compound Object

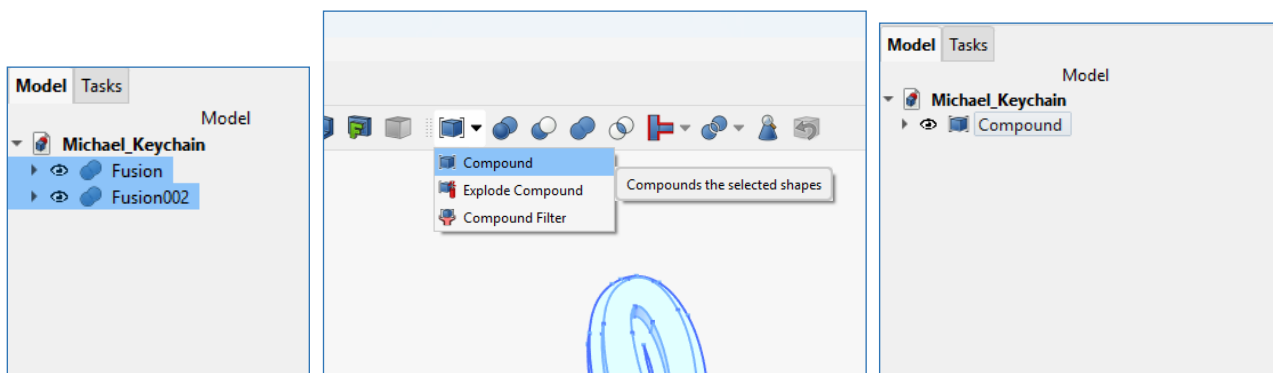
To export both layers as a single file while preserving their individual identities, we use the Compound operation  instead of Union .

Compound vs. Union

If we use Union  here, the two layers will be merged into one solid and the slicer cannot separate them for color assignment. Compound keeps them as separate parts inside one file — exactly what we need for multi-color printing.

1. Hold Ctrl and select both Layer 1 (text fusion) and Layer 2 (background + loop fusion).
2. Go to Part → Compound → Make Compound , or click the Compound icon  in the toolbar.

► Part → Compound → Make Compound 



Compound object created — Model Tree showing 'Compound' containing both fusion objects

5.2 Export as .3MF

1. Select the Compound object in the Model Tree.
2. Go to File → Export or press *Control + E*
3. In the file format dropdown, select 3MF (.3mf).
4. Choose a save location and click Save.

► File → Export → 3MF (.3mf)

5.3 Import into Bambu Studio

1. Open Bambu Studio (or your preferred slicer).
2. Drag and drop the .3MF file onto the slicer window, or use File → Import.

The model will appear as a single object on the build plate.



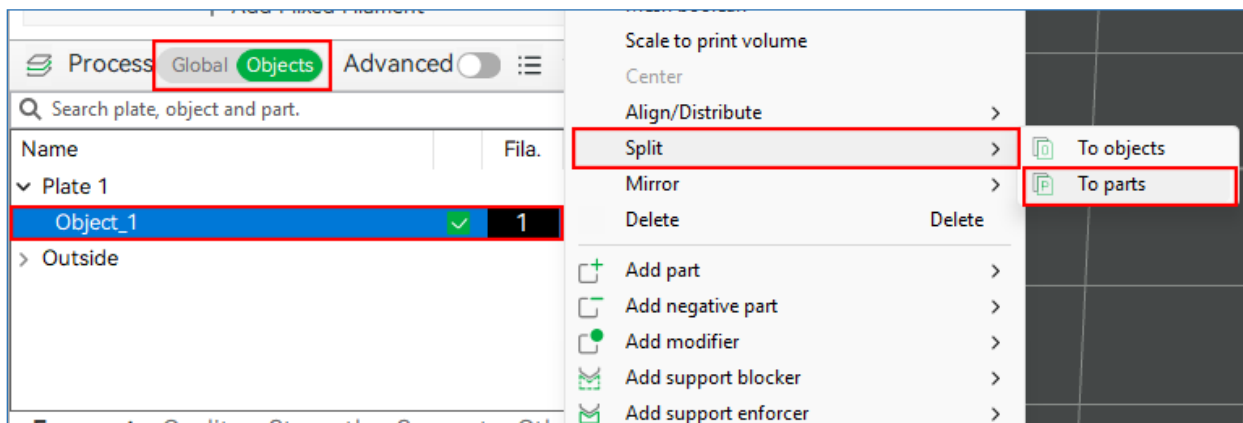
Bambu Studio with the imported .3MF file showing on the build plate

5.4 Split to Parts and Assign Colors

1. In Bambu Studio, make sure you are in Object mode (not Global mode) — check the toggle in the top bar.
2. In the Objects panel (right side), locate Object 1 under Plate 1.
3. Right-click the object name.
4. Go to Split → Split to Parts (not 'Split to Objects').

⚠ Split to Parts — Not Split to Objects

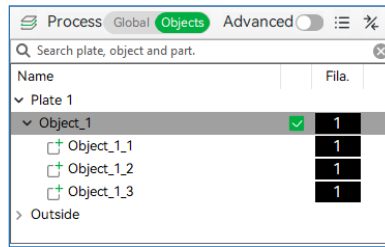
'Split to Parts' keeps all pieces on the same object, making color assignment straightforward. 'Split to Objects' creates separate objects and complicates the workflow.



Right-click menu with 'Split → Split to Parts' highlighted

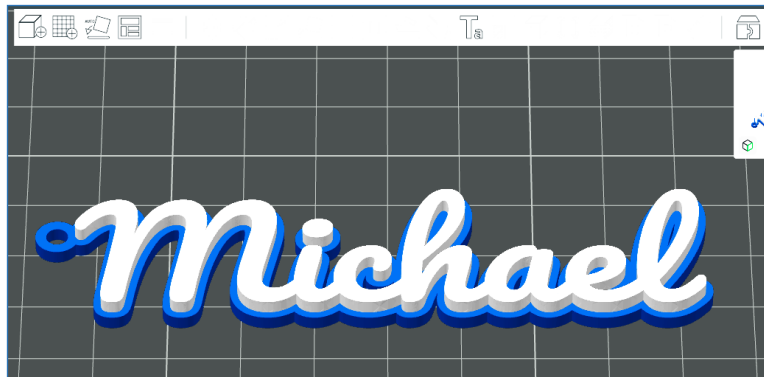
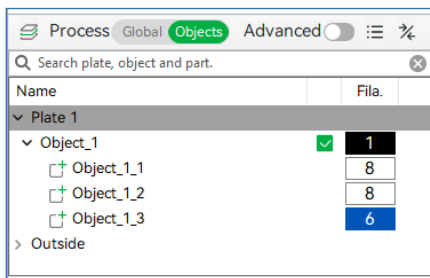
After splitting, you will see multiple parts listed:

- Layer 2 — the background plate + ring loop (bottom layer)
- Layer 1 — the text solid (top layer)
- Small separate elements — e.g., the dot on the letter 'i' (if applicable)



Objects panel after splitting — showing parts listed separately

1. Select all bottom layer parts. If there are multiple, hold Ctrl to multi-select.
2. Right-click → Change Filament → Select your desired color (e.g., white or light color).
3. Select all text layer parts.
4. Right-click → Change Filament → Select a contrasting color (e.g., black, gold, or red).

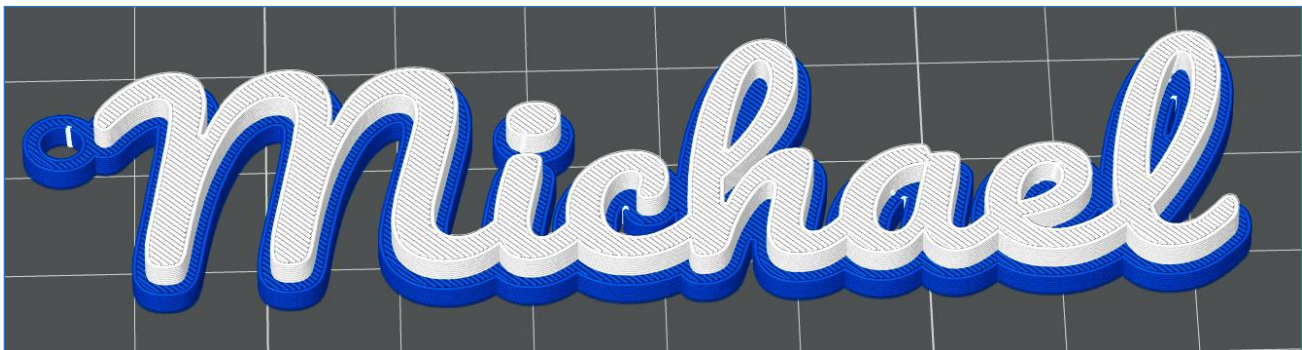


Both layers with different filament colors assigned — visible as different colors on the model

Background (Layer 2)	Suggested: White, Beige, Silver, or any light color
Text (Layer 1)	Suggested: Black, Gold, Red, or any high-contrast color
Ring Loop	Same filament as Layer 2 (it is part of Layer 2)

5.5 Slice and Print

1. Click Slice Now in Bambu Studio.
2. Preview the layer-by-layer result to confirm the two colors are separated correctly.
3. Send the job to your printer.



☒ **Step 5 Complete — Congratulations!**

Your custom multi-color name keychain is now fully modeled, exported, and ready to print. Repeat this workflow with any name, word, or short text to create personalized keychains for friends, events, or retail.



Troubleshooting & Common Issues

Text Shape Issues

Font not loading	Make sure the font file is a valid .ttf or .otf file. Browse to the font using the '...' button in the Shape String dialog.
Letters look wrong	Some decorative fonts have complex curves that may not translate cleanly. Try a simpler font if the preview looks incorrect.
Text appears very small	The height field in Shape String is in millimeters. Increase it (20–30 mm is recommended for keychains).

Extrusion Issues

Extrusion goes in wrong direction	Select the Extrude object, open the Data tab, and enable the Reversed option.
Union fails with an error	Try applying the Union to smaller groups of letters, then union those groups together.
Internal edges visible after union	This is normal cosmetic behavior. It does not affect the 3D print quality.

Offset Issues

Offset shape is broken or self-intersecting	Reduce the offset value (try 0.5 mm instead of 1 mm). Very tight letter spacing can cause issues with large offsets.
FaceBinder shows nothing	Make sure you selected the back faces (not the front or side faces) before creating the FaceBinder.

Multi-Color Printing Issues

Slicer shows only one color	Make sure you used Compound (not Union) in FreeCAD before exporting.
Parts are not separating in slicer	Use 'Split to Parts' — not 'Split to Objects'.
Colors assigned to wrong layers	Check which part is which in the Objects panel. You can click each part and it will highlight in the viewport.

Design Tips & Customization

Text & Font Choices

- Cursive / script fonts produce a more elegant look and create natural letter connections for better structural strength.
- Bold sans-serif fonts (e.g., Impact, Futura Bold) give a clean, modern appearance.
- Avoid very thin or hairline fonts — they may be too fragile to print at small sizes.
- Keep text height between 15–25 mm for typical keychain sizes.

Layer Thickness

Layer 1 (text)	1.5 – 3 mm recommended. Thicker text = more visible in contrast printing.
Layer 2 (background)	1.5 – 3 mm. Can be the same as or slightly thinner than Layer 1.
Total combined	3 – 5 mm total is comfortable for a keychain tag.

Ring Loop Size

Outer diameter	6 mm is suitable for standard keyring wire (~3 mm diameter).
Inner diameter	3 mm minimum. Increase if using thicker keyring hardware.
Overlap with body	Aim for 1–2 mm overlap for a strong connection during printing.

Multi-Layer Designs

The same workflow scales to 3 or more layers. Simply repeat the FaceBinder → Offset → Extrude → Union process for each additional layer, increasing the offset distance by 1 mm per layer (e.g., 1 mm, 2 mm, 3 mm for three layers).

Adding a Logo or Symbol

You can combine this text workflow with an imported SVG logo. Import the SVG via File → Import into the Draft Workbench, then follow the same Extrude and Union steps. Combine the logo solid and the text solid using Union before the Compound step.

Quick Reference — Workflow Summary

Step 1	Draft WB → Drafting → Shape from Text → Enter text, font, height → OK
Step 2	Part WB → Select ShapeString → Part → Extrude (2 mm) → OK → Part → Boolean → Union
Step 3	Rotate view → Select back faces → Draft WB → Face Binder → Part WB → 2D Offset (1 mm, Pipe, Arc) → Extrude (2 mm) → Union
Step 4	Select top face → New Sketch → Sketcher → Two circles (Ø6 mm outer, Ø3 mm inner) → Position → Close → Part WB → Extrude (2 mm) → Union with Layer 2
Step 5	Select both layers → Part → Compound → File → Export as .3MF → Slicer: Import → Split to Parts → Assign Colors → Slice → Print

Support & Resources

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 Ko-fi: <https://ko-fi.com/cadcamcourse>

Your support helps create more detailed, beginner-friendly content for the FreeCAD and 3D printing community.

Subscribe to the channel and turn on notifications to stay updated.

Thank you for following along — happy 3D printing! 🖨️